Transforming SysML v1 (DELS) into Many Small SysML v2 Artifacts

XML → IR → Deterministic v2 → 200–500 line Shards

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https://github.com/usnistgov/DiscreteEventLogisticsSystems

Objective

- Convert large SysML v1 models (`.mdzip`, `.xml`) to high-quality SysML v2 *textual* artifacts.
- Produce **many small, semantically coherent files** (target: 200–500 lines each).
- Generate paired **NL summaries** for each file for a SysMLv2–NL dataset.
- Keep the process **deterministic** and **auditable**; use LLMs only for stylistic polish.

Pipeline Overview

1. **Ingest v1**: unzip `.mdzip`/parse XML; normalize profiles/stereotypes.
2. **Extract IR**: JSON intermediate representation (structure, behavior, requirements).
3. **Rule-based mapping**: deterministic v1→v2 text.
4. **Sharding**: split into 200–500 line cohesive files.
5. **LLM pass**: refine style, not semantics.
6. **Validation**: syntax + semantic checks, imports/exports.
7. **Packaging**: NL summaries + manifest (.jsonl).

Intermediate Representation (IR)

```
{
  "blocks": [{
    "id":"b:WarehouseSystem",
    "name":"WarehouseSystem",
    "parts":[{"name":"conveyor","type":"Conveyor"}],
    "valueProperties":[{"name":"throughput","type":"item/s"}],
    "ports":[{"name":"cmdIn","kind":"proxy","interface":"CmdIF"}]
  }],
  "requirements":[{"id":"1.2","name":"Performance","text":"..."}],
  "activities":[{"name":"InboundShipment","actions":[...]}]
}
```

v1→v2 Mapping Rules

Structure

- Block → `part def`
- Part property → `part` membership
- Value property → `attribute`/constraint
- Connector → `connection`

Requirements

- Requirement → `requirement` def
- Links → v2 allocations/satisfy relations

Behavior

- Activity → `action defs`
- Sequence → occurrences/messages
- Use case → `usecase defs`

Stereotypes

- → v2 `metadata` / `#keywords`

Sharding Strategy (200–500 lines each)

- Split by concern: Structure / Behavior / Requirements / Interfaces.
- Within each concern, anchor on top block or scenario root.
- Greedy BFS add elements until $\sim$400 lines.
- Record `exports`/`imports`.
- Keep large units/enums in shared `Libraries/`.

Example Generated v2 Chunk

```
package Structure::WarehouseSystem;
import Libraries::Units;

part def WarehouseSystem;
WarehouseSystem {
  part conveyor: Conveyor;
  attribute throughput: Real; // unit: item/s
  port cmdIn: In<CmdIF>;
  connection cmdIn -> ControlStation.cmdOut using item Cmd;
}
```

Output Dataset Organization

- Each file = one coherent SysMLv2 chunk (200–500 lines)
- Paired natural-language summary
- **Manifest (.jsonl)** entry:

```
{
 "id":"Structure/Warehouse.CoreParts",
 "sysml2_path":"SysMLv2/Structure/Warehouse.CoreParts.sysml",
 "nl_path":"NL/Structure/Warehouse.CoreParts.md",
 "lines":372,
 "exports":["WarehouseSystem"],
 "imports":["AGV","Pallet"]
}
```

LLM Role (Polish Only)

- Input: deterministic v2 + IR snippet
- Output: stylistically refined v2 text or short NL summary
- Guardrail prompt:
  "Do not invent new elements. Only reorganize or reformat existing definitions for v2 idiom."

Validation & Expert Feedback Points

- Check mapping fidelity vs. v1 semantics.
- Confirm v2 package layout and naming best practices.
- Discuss trade-offs: *satisfy* vs. *allocation usage*.
- Recommend validation checks for consistency across shards.

Summary

- Deterministic pipeline: v1 XML $\rightarrow$ IR $\rightarrow$ v2 $\rightarrow$ 200–500 line shards.
- NL summaries and manifests for dataset creation.
- Ready for expert feedback on mapping conventions.